

Notes: Grinblatt and Keloharju (JF, 2001)
How Distance, Language, and Culture Influence Stockholdings and Trades

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1 Big picture

- **Research question.** Do investors favor familiar firms even within a single country? More specifically, do geographic distance, native language, and cultural proximity influence which stocks investors hold, buy, and sell?
- **Main idea.** Home bias is usually studied across countries. This paper shows that the same familiarity forces operate inside a country, where capital controls, currency differences, and international institutional frictions are largely absent.
- **Empirical setting.** Finland is unusually useful because it has comprehensive investor-level ownership and trading records, two official languages, and measurable geographic and cultural variation across investors and firms.
- **Core finding.** Investors are more likely to hold, buy, and sell stocks of firms that are geographically close, that communicate in the investor’s native language, and that have CEOs from the investor’s cultural background.
- **Sophistication gradient.** The effects are weaker for financially sophisticated institutions and for households with more diversified portfolios, but they do not disappear.
- **Economic takeaway.** Familiarity affects portfolio choice. Investors do not simply diversify across all available securities; they overweight firms that feel close in geography, language, or culture.

2 Incremental contribution of the paper

- **From international home bias to local familiarity bias.** Earlier work documented investors’ reluctance to hold foreign stocks. Grinblatt and Keloharju show that bias toward familiar assets exists even within one national market.
- **Direct ownership and transaction data.** The paper studies actual holdings, open-market purchases, and open-market sales rather than relying only on aggregate portfolio shares or survey evidence.
- **Separate distance, language, and culture.** The Finnish setting allows the authors to distinguish three dimensions of familiarity that are often bundled together in international comparisons.
- **Investor heterogeneity.** The paper compares households, all institutions, financially savvy institutions, and less savvy institutions; it also sorts households by portfolio diversification.
- **Firm fixed effects and municipality controls.** Much of the analysis compares how investors in different municipalities allocate weight across the same firms, helping absorb firm-level differences such as size, visibility, and national importance.
- **Trading as well as ownership.** By examining buys and sells, the paper shows that familiarity is not only a legacy of IPO allocations, employee ownership, or historical holdings.

3 Data and measurement

3.1 Investor and trading data

The main data come from the Finnish Central Securities Depository. The database contains official records of shareownership and trades for Finnish investors.

- **Ownership snapshot.** The ownership data are measured as of January 1, 1997.
- **Trading window.** The transaction data cover purchases and sales from December 27, 1994, to January 10, 1997.
- **Firms.** The full set contains 97 publicly traded Finnish firms; the shareownership analysis uses 92 firms after excluding a very small firm and firms delisted before the ownership snapshot.
- **Investor type.** Investors are classified as households or institutions. Institutions are further divided into more financially savvy institutions and less savvy institutions.
- **Aggregation.** Because the raw investor-level data are extremely large, the analysis aggregates investors to municipality-firm cells. Each cell records ownership and trading weights for a given investor group in a given municipality and firm.

Interpretation. The data allow the paper to compare, for example, whether investors in one municipality overweight the nearby firm relative to investors elsewhere in Finland.

3.2 Distance

Distance is measured between the centroid of the investor’s municipality and the centroid of the municipality of the firm’s headquarters.

- If an investor and a firm are in the same municipality, distance is defined as one-quarter of the distance to the nearest neighboring municipality.
- The analysis distinguishes firms headquartered in the greater Helsinki area from firms headquartered elsewhere.
- The distance effect is allowed to change after 100 kilometers, because the relationship is strongest at short distances and flattens at longer distances.

Interpretation. Distance is not merely a physical measure. It proxies for local information, local visibility, social interaction, media coverage, customer familiarity, and possibly employee or supplier relationships.

3.3 Language

Finland has two official languages, Finnish and Swedish. Finnish speakers are the large majority of the population, but Swedish-speaking investors hold a disproportionately large share of household equity wealth.

Firm language is measured by the language of the firm’s annual report:

- Swedish-only annual reports;
- Finnish-only annual reports;
- multilingual annual reports.

Interpretation. Annual report language is a proxy for the language in which the firm communicates with investors. It can matter directly through information access and indirectly through identity or familiarity.

3.4 Culture

Firm culture is proxied by the cultural origin of the CEO. The authors classify CEOs as Finnish or Swedish cultural origin using names and native-language information.

- Most sample firms have CEOs of Finnish cultural origin.
- A minority have CEOs of Swedish cultural origin.
- Culture is related to, but distinct from, the language of the annual report.

Interpretation. The culture variable asks whether investors prefer firms whose senior management appears culturally close to them, even after controlling for distance and the language used in investor communications.

3.5 Investor sophistication

The paper examines whether familiarity effects are weaker for more sophisticated investors.

- Institutions are separated into financially savvy institutions and less savvy institutions.
- Households are separated by the number of distinct stocks in their portfolios.
- More diversified household portfolios are treated as a proxy for greater investment sophistication or lower familiarity bias.

4 Methodology and empirical specifications

4.1 Simple ratio evidence

The paper first computes transparent ratios that compare a firm’s weight among one investor group with the firm’s weight among all investors in Finland.

For the distance effect, the ratio is:

$$R_i^{\text{distance}} = \frac{\text{Firm } i\text{'s weight among investors in the headquarters municipality}}{\text{Firm } i\text{'s weight among all investors in Finland}}.$$

For language and culture, analogous ratios compare the firm’s weight among Swedish-speaking investors with its weight among all investors.

- A ratio above one means that the investor group overweights the firm.
- A ratio below one means that the investor group underweights the firm.
- The ratios are computed separately for holdings, buys, and sells, and separately for households and institutions.

4.2 Multivariate firm-municipality regressions

The main regression framework uses firm-municipality observations. The dependent variable is the difference between a firm’s weight in a municipality and the firm’s overall weight in Finland:

$$Y_{ij} = w_{ij} - w_i,$$

where w_{ij} is firm i ’s weight among investors in municipality j , and w_i is firm i ’s weight among all investors in Finland.

A simplified version of the specification is:

$$Y_{ij} = \alpha + f(d_{ij}) + \lambda_i + \mu_j + \varepsilon_{ij},$$

where $f(d_{ij})$ captures distance from municipality j to firm i , λ_i are firm effects, and μ_j are municipality effects.

Interpretation. Firm effects absorb general firm popularity, size, liquidity, and national prominence. Municipality effects absorb differences in how much each municipality invests overall.

4.3 Piecewise distance specification

The distance specification is piecewise in the log of distance:

$$f(d_{ij}) = \beta_1 \min\{\ln 100, \ln d_{ij}\} + \beta_2 \max\{\ln 100, \ln d_{ij}\} + \text{Helsinki interactions}.$$

This allows the slope of the distance effect to change after 100 kilometers and allows the effect to differ for firms headquartered in the greater Helsinki area.

- The paper finds that the distance effect is strongest at short distances.
- The effect is weaker for firms headquartered in the Helsinki area, which are more nationally visible.
- Figure 1 validates the piecewise specification by plotting five-kilometer distance dummy coefficients.

4.4 Language and culture in the regression framework

To separate distance from language and culture, the paper includes variables that interact the fraction of Swedish-speaking investors in a municipality with firm language and firm culture:

$$\begin{aligned} Y_{ij} = & \alpha + f(d_{ij}) + \theta_1(\text{SwedishShare}_j \times \text{FinnishReport}_i) \\ & + \theta_2(\text{SwedishShare}_j \times \text{SwedishReport}_i) + \theta_3(\text{SwedishShare}_j \times \text{SwedishCEO}_i) \\ & + \lambda_i + \mu_j + \varepsilon_{ij}. \end{aligned}$$

Interpretation. If Swedish-speaking investors prefer Swedish-language or Swedish-culture firms, the relevant interaction coefficients should be positive.

4.5 Portfolio-diversification tests

The paper repeats the ownership regressions for household investors grouped by the number of stocks in their portfolios.

- Investors with one or two stocks are likely to be less diversified and possibly less sophisticated.
- Investors with many stocks are more diversified and likely less affected by local or cultural familiarity.
- The key test is whether distance, language, and culture effects decline as the number of stocks in the household portfolio rises.

5 Identification and interpretation

5.1 What the paper can identify

The paper identifies strong cross-sectional patterns in ownership and trading: investors overweight nearby, linguistically familiar, and culturally familiar firms. The evidence is especially strong because the analysis is within one country and uses firm and municipality controls.

5.2 Why Finland is a powerful setting

Finland offers three advantages.

- A centralized depository gives detailed ownership and trading data.
- Two official languages create variation in investor language and firm communication language.
- Municipal geography allows precise measurement of investor-firm distance.

This setting avoids many confounds in international home-bias studies, such as capital controls, tax treaties, currency differences, and foreign listing constraints.

5.3 Employee and customer concern

A concern is that investors near a firm may be employees, suppliers, or customers rather than ordinary investors exhibiting familiarity bias. The paper addresses this concern by analyzing open-market buys and sells, not only ownership, and by checking robustness after excluding investors in the firm's own municipality.

5.4 Language versus culture

Language and culture are correlated, but the paper separates them by using both annual-report language and CEO cultural origin. Table IV is especially important because it studies the fraction of shareholders who speak Swedish and shows that both language and CEO culture matter.

5.5 Sophistication interpretation

The evidence that familiarity effects are weaker among financially savvy institutions and more diversified households suggests that the patterns are not purely mechanical. Less sophisticated investors appear more influenced by familiarity, although sophisticated investors are not completely immune.

5.6 Limitations

The analysis is observational. Distance, language, and culture may proxy for information, trust, attention, local media exposure, social networks, or taste-based identity preferences. The paper documents the empirical patterns convincingly, but it cannot fully distinguish all underlying mechanisms.

6 Tables and figures

6.1 Table I: Simple evidence on distance, language, and culture

Read:

- Panel A reports ratios of firm weights among investors in the headquarters municipality relative to firm weights among all investors in Finland.
- The distance effect is large. For non-Helsinki firms, the median household shareownership ratio is 12.16 and the institutional ratio is 8.12.
- For Helsinki-area firms, the distance effect is still present but much smaller, consistent with Helsinki firms being more nationally known.
- Panels B and C show that Swedish-speaking investors overweight Swedish-language and Swedish-culture firms and underweight Finnish-only firms.
- The effects appear in ownership, purchases, and sales, suggesting that they are not only inherited from IPO allocations or old holdings.

Economic message: Even before regression controls, investors overweight geographically, linguistically, and culturally familiar firms. Familiarity is visible in both static portfolios and active trading.

Table I
Summary Statistics Documenting the Influence of Distance, Language, and Culture

Panel A reports the median and fraction exceeding one on the ratio of firm i 's weight for investors in the municipality of its headquarters over firm i 's weight among all investors in Finland. Panels B and C report the median and fraction exceeding one statistics on the ratio of firm i 's weight among Swedish-speaking investors over firm i 's weight among all investors. Panel B focuses on subsamples of firms generated by the language of the firm, and Panel C's subsamples are generated by the cultural origin of the CEO. Weights are based on the number of shareowners, buys, and sells, and are separately computed for households, institutions, and for Panel A, both greater Helsinki area headquartered firms and non-Helsinki firms. The number of shareowners is computed as of January 1, 1997, whereas the buys and sells are computed between December 27, 1994, and January 10, 1997.

	Weights Based on Number of					
	Shareowners		Buys		Sells	
	Households	All Institutions	Households	All Institutions	Households	All Institutions
Panel A: Distance Effect						
Summary Statistics for the Ratio <i>Numerator/Denominator</i> <i>Numerator</i> = Firm i 's Weight among Investors in its Municipality <i>Denominator</i> = Firm i 's Weight among All Investors in Finland						
Median for firms of following type						
Helsinki area headquartered companies ($n = 62$)	1.41	1.26	1.11	1.00	1.11	0.95
Rest of Finland headquartered companies ($n = 35$)	12.16	8.12	4.22	2.11	9.95	14.75
All companies ($n = 97$)	1.81	1.45	1.34	1.01	1.31	1.06
Fraction greater than 1 for firms of following type						
Helsinki area headquartered companies	0.83	0.78	0.71	0.47	0.69	0.42
Standard error	0.07	0.07	0.06	0.06	0.06	0.06
Rest of Finland headquartered companies	1.00	0.91	0.89	0.53	0.94	0.89
Standard error	0.09	0.09	0.08	0.08	0.08	0.08
All companies	0.89	0.83	0.77	0.53	0.78	0.59
Standard error	0.05	0.05	0.05	0.05	0.05	0.05

Table I, Panel A: Distance effect.

Panel B: Language Effect						
Summary Statistics for the Ratio <i>Numerator/Denominator</i> <i>Numerator</i> = Firm i 's Weight among Investors who Speak Swedish <i>Denominator</i> = Firm i 's Weight among All Investors in Finland						
Median for firms of following type						
Annual report only in Swedish $n = 2$	7.77	4.12	4.84	3.13	4.95	2.65
Annual report only in Finnish $n = 12$	0.52	0.43	0.59	0.61	0.38	0.56
Annual report multilingual $n = 83$	0.85	0.98	0.92	1.09	0.89	0.98
Fraction greater than 1 for firms of following type						
Annual report only in Swedish	1.00	1.00	1.00	1.00	1.00	1.00
Standard error	0.35	0.35	0.35	0.35	0.35	0.35
Annual report only in Finnish	0.00	0.10	0.17	0.25	0.00	0.17
Standard error	0.16	0.16	0.14	0.14	0.14	0.14
Annual report multilingual	0.35	0.46	0.43	0.60	0.41	0.47
Standard error	0.06	0.06	0.05	0.05	0.05	0.05
Panel C: Culture Effect						
Summary Statistics for the Ratio <i>Numerator/Denominator</i> <i>Numerator</i> = Firm i 's Weight among Investors who Speak Swedish <i>Denominator</i> = Firm i 's Weight among All Investors in Finland						
Median for firms of following type						
CEO Swedish culture $n = 14$	2.49	1.82	1.73	1.34	2.13	1.15
CEO Finnish culture $n = 83$	0.77	0.87	0.86	1.00	0.80	0.94
Fraction greater than 1 for firms of following type						
CEO Swedish culture	0.71	0.79	0.79	0.93	0.79	0.57
Standard error	0.13	0.13	0.13	0.13	0.13	0.13
CEO Finnish culture	0.27	0.38	0.35	0.51	0.30	0.42
Standard error	0.06	0.06	0.05	0.05	0.05	0.05

Table I, Panels B and C: Language and culture effects.

6.2 Table II and Figure 1: Distance effects in multivariate regressions

Read Table II:

- Table II estimates firm-municipality regressions for holdings, buys, and sells.
- The dependent variable is the firm's local portfolio weight minus its national weight.
- The regressions include firm and municipality dummy variables, so the distance effect is not just a firm-size or municipality-wealth effect.
- The distance coefficients show that local investors overweight nearby firms, especially within short distances.
- Language and culture controls remain meaningful, especially for household ownership.

Read Figure 1:

- Figure 1 replaces the piecewise log-distance terms with five-kilometer distance dummies.
- The coefficient is highest at very short distances and declines rapidly with distance.
- The distance gradient is steeper for non-Helsinki firms and weaker for Helsinki-area firms.
- Household investors display a stronger distance pattern than institutional investors.

Economic message: The regression evidence confirms that distance matters after controlling for firm and municipality characteristics. The effect is local, nonlinear, and weaker for more nationally visible firms.

Table II

Multivariate Regressions to Separate the Influence of Distance from that of Language and Culture

Table II reports goodness of fit, number of observations, and coefficients (with *t*-statistics below) for a constant and 7 regressors in each of 12 regressions with the dependent variable being firm *i*'s weight for investors in a given municipality less firm *i*'s weight among all investors in Finland. Weights are based on the number of shareowners, buys, and sells, and are separately computed for households, institutions, survey institutions, and unsurvey institutions. Each data point is a firm-municipality combination. The table does not report the regression coefficients on about 500 other included regressors—dummy variables for the firms and municipalities. The number of shareowners is computed as of January 1, 1997, and the buys and sells are computed between December 27, 1994, and January 10, 1997.

Independent Variables	Dependent Variable: Weight on Firm <i>i</i> in Municipality <i>j</i> Less Overall Weight on Firm <i>i</i> with Weight Computed from											
	# of owners				# of buys				# of sells			
	Institutions			Households	Institutions			Households	Institutions			Households
	Non-Fin. Corp. & Govt. Fin. and Ins. Inst.	Non-Fin. Corp. & Govt. Fin. and Ins. Inst.	Non-Fin. Corp. & Govt. Fin. and Ins. Inst.		Non-Fin. Corp. & Govt. Fin. and Ins. Inst.	Non-Fin. Corp. & Govt. Fin. and Ins. Inst.	Non-Fin. Corp. & Govt. Fin. and Ins. Inst.		Non-Fin. Corp. & Govt. Fin. and Ins. Inst.	Non-Fin. Corp. & Govt. Fin. and Ins. Inst.	Non-Fin. Corp. & Govt. Fin. and Ins. Inst.	
Constant	0.078	0.028	0.014	0.056	0.041	-0.134	-0.132	-0.009	0.040	-0.084	-0.102	0.027
	9.74	1.90	0.87	2.49	3.51	-6.99	-5.88	-0.22	3.36	-4.04	-4.91	0.49
Min [ln 100, ln (distance)]	-0.004	-0.016	-0.014	-0.023	-0.012	-0.012	-0.010	-0.020	-0.028	-0.020	-0.015	-0.004
	-0.02	-1.69	-1.16	-13.16	-14.42	-7.07	-6.00	-0.41	-0.92	-12.86	-7.77	-11.38
Max [ln 100, ln (distance)]	-0.006	-0.003	-0.002	-0.006	-0.003	0.000	0.000	0.007	-0.005	-0.004	-0.003	-0.007
	-0.03	-0.16	-0.12	-0.44	-0.16	0.00	0.00	0.12	-0.04	-0.15	-0.10	-0.28
Min [ln 100, ln (distance)] × Dummy for company headquartered in Greater Helsinki Area	0.021	0.015	0.013	0.020	0.010	0.011	0.009	0.006	0.022	0.015	0.012	0.003
	24.32	9.05	7.54	17.9	8.38	4.41	3.77	0.69	17.42	9.14	8.21	0.95
Max [ln 100, ln (distance)] × Dummy for company headquartered in Greater Helsinki Area	0.002	0.001	0.001	0.003	0.002	0.000	0.000	-0.005	0.005	0.003	0.003	0.006
	0.49	1.23	0.99	1.66	2.72	0.08	0.04	-1.22	8.25	2.79	2.18	2.51
Greater Helsinki Area												
Fraction of Swedish speakers in municipality × Annual report only in Finnish dummy	0.001	-0.001	-0.001	-0.003	0.001	0.000	0.000	-0.001	-0.003	-0.005	-0.003	-0.001
	0.09	-0.67	-0.50	-0.77	0.57	-0.07	-0.07	-0.16	-1.80	-1.19	-1.18	-0.62
Fraction of Swedish speakers in municipality × Annual report only in Swedish dummy	0.011	0.011	0.019	-0.014	0.009	-0.002	-0.001	-0.005	-0.001	0.010	0.014	-0.004
	2.04	2.02	3.53	-1.73	2.34	-0.24	-0.14	-0.07	-0.21	3.27	3.61	-1.01
Fraction of Swedish speakers in municipality × CEO Swedish culture dummy	0.001	0.007	0.006	0.012	0.014	0.003	0.004	-0.007	0.009	0.002	0.001	0.021
	0.19	2.78	2.39	1.07	2.88	0.73	0.73	-0.19	2.49	0.60	0.29	1.03
Adjusted R ²	0.175	0.142	0.112	0.203	0.097	0.222	0.234	0.016	0.077	0.082	0.085	0.079
N	43076	43210	40572	30246	42747	27249	27054	7275	43844	27054	26672	14841

Table II: Distance, language, and culture in firm-municipality regressions.

6.3 Table III: Portfolio diversification and familiarity effects

Read:

- Table III repeats the shareownership regressions for households grouped by the number of stocks in their portfolios.
- Households with very concentrated portfolios show the strongest distance, language, and culture effects.
- The distance effect weakens as portfolio diversification rises.
- The language and culture variables remain positive in many more-diversified groups, but their magnitudes are generally smaller.

Economic message: Familiarity bias is strongest among less diversified households, consistent with the view that investor sophistication reduces but does not eliminate local and cultural preferences.

6.4 Table IV: Language and culture of shareholders

Read:

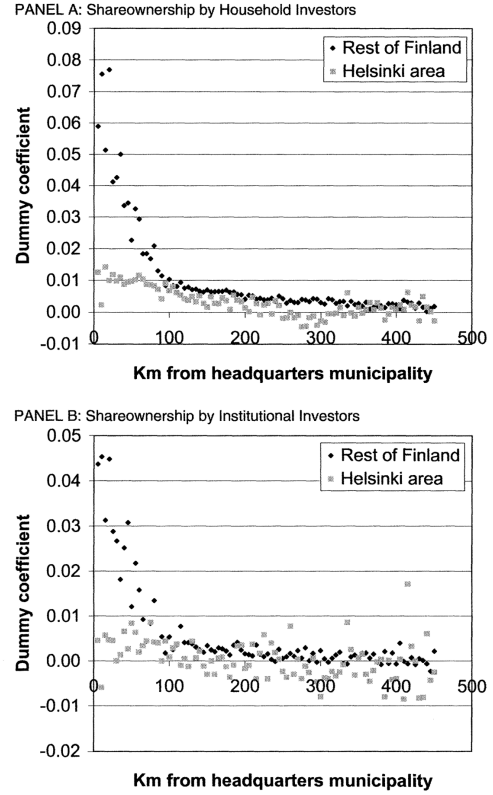


Figure 1: Distance dummy coefficients for household and institutional shareownership.

- Table IV changes the dependent variable to the fraction of a firm’s shareholders in a municipality who speak Swedish.
- Swedish-speaking investors are less likely to hold Finnish-only annual-report firms and more likely to hold Swedish-only annual-report firms.
- A Swedish-culture CEO is also associated with a higher Swedish-speaking shareholder share.
- The CEO culture coefficient is positive across households and institutions.
- The household results are especially strong, suggesting that language and culture are powerful determinants of retail investor familiarity.

Economic message: Language and culture are not just reflections of geography. Even after controlling for distance and municipality effects, investors sort into firms that communicate in their language and have culturally similar leadership.

Table IV
Multivariate Regressions that Analyze the Determinants of the Language and Culture of Investors in a Firm–Municipality Combination

Table IV reports goodness of fit, number of observations, and coefficients (with *t*-statistics below) for a constant and eight regressors. The dependent variable is the fraction of Swedish-speaking shareowners. Unreported dummies for each municipality are also included. The regressions are separately computed for households, institutions, savvy institutions, and unsavvy institutions. Each data point is a firm–municipality combination. The number of shareowners is computed as of January 1, 1997.

Independent Variables	Dependent Variable: The Fraction of Firm <i>i</i> 's Shareowners Residing in Municipality <i>j</i> who Speak Swedish			
	Households	Institutions		
		All	Non-Fin. Corp & Fin. and Ins. Inst.	Government & Nonprofit Institutions
Constant	0.067 1.55	0.097 1.26	0.075 0.92	0.105 0.60
Min [ln 100, ln (distance)]	0.005 1.31	0.008 1.31	0.006 0.90	0.018 1.72
Max [ln 100, ln (distance)]	−0.009 −2.98	−0.016 −2.38	−0.012 −1.68	−0.025 −1.48
Min [ln 100, ln (distance)] × Dummy for company headquartered in Greater Helsinki Area	0.005 0.94	−0.005 −0.56	−0.002 −0.22	−0.019 −1.37
Max [ln 100, ln (distance)] × Dummy for company headquartered in Greater Helsinki Area	0.011 3.04	0.010 1.24	0.012 1.33	−0.005 −0.23
Dummy for company headquartered in Greater Helsinki Area	−0.193 −3.32	−0.068 −0.68	−0.122 −1.14	0.278 1.22
Annual report only in Finnish dummy	−0.020 −6.43	−0.011 −1.70	−0.005 −0.73	−0.065 −3.13
Annual report only in Swedish dummy	0.062 8.29	0.009 0.52	0.020 1.07	0.063 1.34
CEO Swedish culture dummy	0.025 11.32	0.032 6.76	0.025 4.87	0.035 3.83
Adjusted <i>R</i> ²	0.834	0.824	0.817	0.770
<i>N</i>	21279	7612	6928	2521

Table III
Multivariate Regressions that Analyze the Relation between Portfolio Diversification, and the Influence of Distance, Language, and Culture

Table III reports goodness of fit, number of observations, and coefficients (with *t*-statistics below) for a constant and seven regressors in each of 10 regressions with the dependent variable being firm *i*'s shareowner weight for investors in a given municipality with a specified number of distinct stocks in their portfolio less firm *i*'s shareowner weight among all investors in Finland with the same number of stocks. Each data point is a firm–municipality combination. The table does not report the regression coefficients on about 550 other included regressors—dummy variables for the firms and municipalities. The number of shareowners is computed as of January 1, 1997.

Dependent Variable: Weight on Firm <i>i</i> in Municipality <i>j</i> Less Overall Weight on Firm <i>i</i> with Weight Computed from # of Owners	Number of Stocks in Portfolio									
Independent Variables	1	2	3	4	5	6	7	8	9	>9
Constant	0.068	0.113	0.097	0.091	0.074	0.073	0.066	0.062	0.061	0.038
Min [ln 100, ln (distance)]	0.06	9.64	8.95	8.31	6.76	6.73	5.85	5.48	5.35	4.98
Max [ln 100, ln (distance)]	−0.022	−0.034	−0.027	−0.026	−0.022	−0.017	−0.017	−0.014	−0.011	−0.008
Min [ln 100, ln (distance)] × Dummy for company headquartered in Greater Helsinki Area	−26.88	−39.41	−33.84	−31.96	−27.08	−21.08	−19.67	−16.91	−12.36	−13.65
Max [ln 100, ln (distance)] × Dummy for company headquartered in Greater Helsinki Area	−0.004	−0.008	−0.008	−0.007	−0.006	−0.006	−0.003	−0.004	−0.004	−0.004
Min [ln 100, ln (distance)] × Dummy for company headquartered in Greater Helsinki Area	−7.53	−13.38	−13.46	−12.79	−10.39	−9.67	−7.73	−6.58	−6.46	−8.16
Max [ln 100, ln (distance)] × Dummy for company headquartered in Greater Helsinki Area	0.020	0.030	0.023	0.022	0.019	0.015	0.013	0.012	0.010	0.007
Fraction of Swedish speakers in municipality × Annual report only in Finnish dummy	16.36	23.31	19.72	18.67	15.82	12.78	11.99	10.13	8.21	8.11
Fraction of Swedish speakers in municipality × Annual report only in Swedish dummy	0.001	0.002	0.003	0.003	0.002	0.001	0.002	0.001	−0.001	0.001
Fraction of Swedish speakers in municipality × CEO Swedish culture dummy	1.09	3.64	4.47	4.39	3.85	1.53	2.47	0.83	−0.76	2.43
Adjusted <i>R</i> ²	0.002	0.000	−0.001	0.001	0.001	0.001	0.001	0.000	0.000	0.000
<i>N</i>	1.52	−0.27	−0.41	0.35	0.41	0.54	0.98	0.32	0.21	−0.03
	0.055	−0.034	0.007	0.023	0.033	0.024	0.037	0.030	0.029	0.017
	14.19	−8.54	1.80	6.37	8.96	6.35	10.24	8.38	7.49	5.95
	0.012	0.003	0.019	0.019	0.016	0.015	0.015	0.010	0.010	0.009
	7.81	21.04	13.07	13.25	10.83	10.15	10.01	7.16	6.22	8.12
	0.275	0.137	0.113	0.087	0.064	0.049	0.048	0.031	0.031	0.035
	41976	41676	41485	40296	38916	34468	32394	30452	25300	22476

Table III: Familiarity effects by household portfolio diversification.

Table IV: Language and culture determinants of Swedish-speaking shareholder shares.

7 Short summary

- The paper studies whether familiarity affects portfolio choice within Finland.
- It uses comprehensive Finnish ownership and trading records around 1995–1997.
- Investors prefer firms that are geographically close to them.
- The distance effect is strongest over short distances and weaker for firms headquartered in the Helsinki area.

- Investors also prefer firms that communicate in their native language.
- Swedish-speaking investors overweight Swedish-language and Swedish-culture firms.
- CEO cultural origin matters even after accounting for firm language and investor-firm distance.
- Familiarity effects are weaker among financially savvy institutions and more diversified households.
- The paper therefore provides within-country evidence that home bias is part of a broader familiarity bias.

8 Conclusion

8.1 Core conclusion

Grinblatt and Keloharju show that distance, language, and culture influence investors' stockholdings and trades. Investors are more likely to hold, buy, and sell stocks of nearby firms, firms that communicate in their native language, and firms whose CEOs share their cultural background.

8.2 Economic intuition

Familiar firms may feel easier to understand, easier to monitor, or more trustworthy. Local investors may encounter more news, social discussion, customer experience, or informal information about nearby firms. Language lowers information-processing costs. Cultural similarity may increase trust or salience.

8.3 Contribution revisited

The key contribution is to make home bias more granular. The paper shows that investor bias toward familiar assets does not require national borders. It appears inside a single integrated market and operates through distance, language, and culture.

8.4 Implications

The findings imply that investor attention and familiarity can shape asset demand. Firms may care about investor relations in local and language communities. Researchers studying portfolio choice should treat geography, communication language, and identity as economically relevant variables.

8.5 Limitations

The evidence does not prove whether familiarity works through superior information, psychological comfort, trust, local loyalty, or some combination. The Finnish setting is uniquely clean, but it may also have features that differ from larger or less linguistically segmented markets.

8.6 Takeaway

The enduring takeaway is that investors do not allocate capital only according to risk and return. They also allocate capital toward what they know, understand, and identify with. Distance, language, and culture are therefore real determinants of financial behavior.